ALGORITHM

Spring 2026, Quiz 5, April 1

11:45-12:05 am

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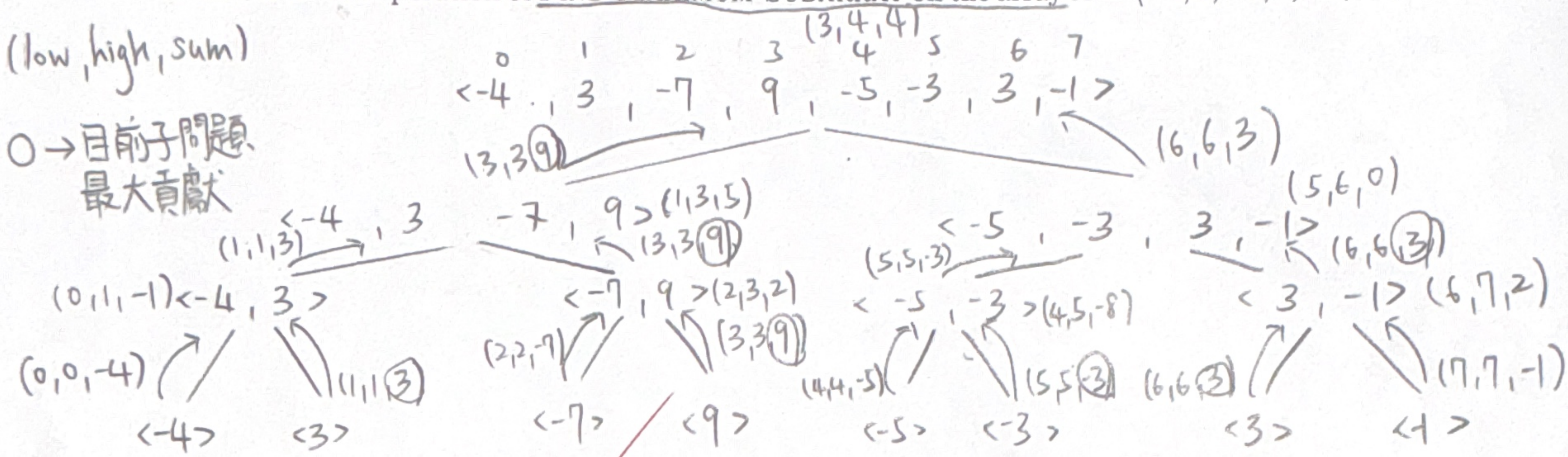
1. Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -4, 3, -7, 9, -5, -3, 3, -1 \rangle$. (8)



max-subarray = 9 #

2. Use the master method to give tight asymptotic bounds for the following recurrences.

(a) $T(n) = 64T(n/4) + n$.

(b) $T(n) = 64T(n/4) + n^3$.

(c) $T(n) = 64T(n/4) + n^5$.

< $T(n) = aT(\frac{n}{b}) + f(n)$, compare $n^{\log_b a}$ 和 $f(n)$ 大小，还有 ε >
< 正規條件 $af(\frac{n}{b}) \le cf(n)$, $c<1$ >
case123 > = <

(a)
$T(n) = aT(\frac{n}{b}) + f(n)$
$= 64T(\frac{n}{4}) + n$
$a=64$, $b=4$ $f(n)=n$
$n^{\log_b a} = n^{\log_4 64} = n^3 > f(n) = n$,
且 $\varepsilon = 3-1 = 2 > 0$
$\therefore$ by Case 1, $T(n) = \theta(n^3)$ #

(b)
$T(n) = aT(\frac{n}{b}) + f(n)$
$= 64(\frac{n}{4}) + n^3$
$a=64$, $b=4$, $f(n)=n^3$
$n^{\log_b a} = n^{\log_4 64} = n^3 = f(n) = n^3$
$\therefore$ by Case 2, $T(n) = \theta(n^3 \lg n)$

(c) $T(n) = aT(\frac{n}{b}) + f(n) = 64T(\frac{n}{4}) + n^5$, $a=64$, $b=4$, $f(n)=n^5$
$n^{\log_b a} = n^{\log_4 64} = n^3 < f(n) = n^5$. $\underline{\varepsilon = 5-3 = 2 > 0}$ ①

② $64f(\frac{n}{4}) \le cn^5$
$64(\frac{n}{4})^5 \le cn^5$
$= 4^3 \times (\frac{n^5}{4^5}) \le cn^5$
$= \frac{1}{16}n^5 \le cn^5$
$c \ge \frac{1}{16}$ $(c<1)$
滿足 Regularity Condition

$\because$ 滿足條件 ①.②
$\therefore$ by Case 3, $T(n) = \theta(n^5)$